

Introduction to Agent-Based Models

Steven Hoekstra^{1 2}

¹CeNDEF, University of Amsterdam, ²Tinbergen Institute



UNIVERSITY OF AMSTERDAM
Amsterdam School of Economics

Overview

1 Introduction

2 ABMs in micro

3 ABMs in Finance

4 ABMs in Macro

Why Agent-Based Models?

- Traditional models often revert to a steady state after a shock.
- ABMs can generate **persistent dynamics** and **path dependence**.
- This makes them suitable for forecasting and policy analysis.

Conditional forecast would have pointed to significant upside risk to inflation surge

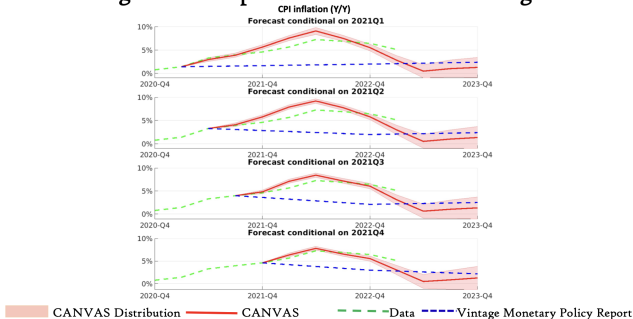


Figure: ABM inflation forecast vs traditional models. (Poledna, 2023)

ABMs vs. DSGE Models

DSGE (traditional)

- Representative agent
- Rational expectations
- Equilibrium dynamics
- Analytical / linearised solutions
- Exogenous shocks drive cycles

Agent-Based Models

- Heterogeneous agents
- Bounded rationality
- Out-of-equilibrium dynamics
- Computational simulation
- Endogenous crises can emerge

ABMs and DSGE are **complementary**: DSGE excels at analytical tractability, ABMs provide flexibility to model heterogeneity, networks, and non-linear feedbacks.

What are Agent Based Models (ABMs)?

- **simulation-based modelling** approach
- **bottoms-up approach** instead of formula-based top-down approach.
- We can analyse systems, perform policy analysis and forecast with ABMs.
- ABMs can give insights which traditional models cannot.
 - Forecast inflation distributions per sector.
 - Stress test the **entire EU banking system** with a 1-1 ABM.
 - Estimate the flood risk per firm/household for an entire country.
- Widely applied in biology, physics, chemistry, sociology, and economics.

What are agent-based model typical elements?

- Multiple agents
- In economics: households, firms, banks, government, and central bank.
- **Local** and/or **global** interaction
- Agents make decisions based on their state/information.
- Interaction based on **topology or network**.
- Environment in which agents are embedded in.
- Global (macro) dynamics are **emergent** from local (micro) interactions.
- Expectations are often **boundedly rational**.
- Conway's Game of Life

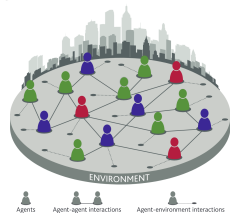


Figure: A schematic of the elements of an agent-based model and their interactions. Source: Haldane and Turrell (2018).

Classic example of agent-based model of Schelling (1969)

- **Agent:** Household/consumer/person.
- **State:** Happiness/utility of the agents.
- **Action:** If neighbours are not similar enough, the agent moves to an empty spot.
- **Topology:** Agents interact only with their direct neighbours.
- **Environment:** The grid where the agents live. In this case, it is rectangular grid.
- Segregation is a **global phenomenon which emerges from local interactions.**
- Example: Demonstration

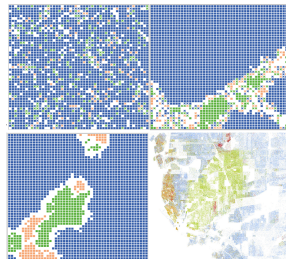


Figure: One of the first economic agent-based model of Schelling (1969). (Source (Axtell and Farmer, Forthcoming))

Simple market in microeconomics

- There are two parties: Many consumers and suppliers.
- The market clears where the upward-sloping supply curve and downward-sloping demand curve cross.
- It is assumed that some auctioneer has all the information to set a price such that supply equals demand.
- But how does the auctioneer know this?
- In fact, this is extremely difficult! The consumer problem has NP complexity (Gilboa et al., 2021).
- Where do we see this in real life?
- Note: Some stylized ABMs like Brock and Hommes (1997) also assume market equilibrium.

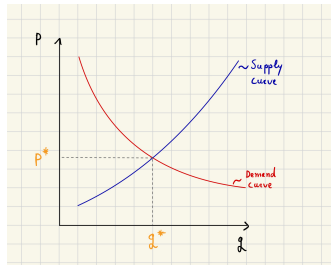


Figure: Microeconomic equilibrium

Double Auction Markets

- Buyers submit bids and sellers submit asks simultaneously based on their valuations.
- A transaction occurs when the bid prices meet or exceed the ask prices.
- No need for an auctioneer.
- Different auction mechanisms also exist, such as:
 - Dutch flower markets
 - English auctions
 - Sealed bid auctions
 - Over-the-Counter (OTC) markets
- Simulation example

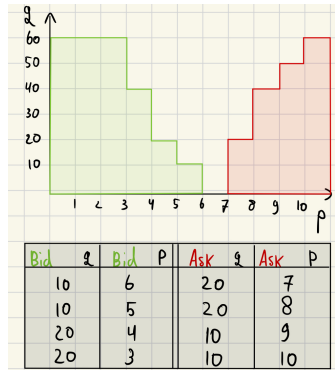


Figure: Double Auction with bid

Decentralized Markets

- Markets can be **decentralized**.
- Buyers or consumers **search** for firms.
- When a firm is found, whether a sale occurs depends on the price and the availability of the good.
- This is called a **Search-and-Matching** mechanism.
- This generates a **bipartite network**
- The seller-buyer network emerges from individual actions.
- Supply does not need to match demand here.

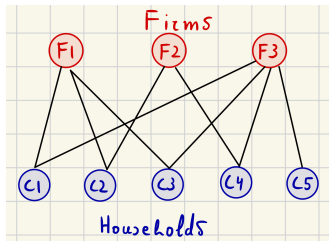
[Go Back](#)

Figure: Decentralized market generated by a search-and-matching mechanism.

Agent-Based Models (ABMs) in Finance

- ABMs can be used to model financial markets and systems.
- We have already seen the double auction market.
- ABMs help explain qualitative financial regularities.
- For example, Brock and Hommes (1997) developed a stylized agent-based model that:
 - Captured stable asset prices.
 - Modeled endogenous bubbles and crashes.
 - Explained clustered volatility.

Banks and Systemic Risk

- Eisenberg and Noe (2001) model financial contagion in an interbank network of debt obligations.
- The model represents banks as nodes with obligations to others and operating cash flows.
- It determines a unique "clearing payment vector" that satisfies bankruptcy rules.
- A single bank default can trigger a cascade of defaults through the network.
- The model is static, analyzing a single clearing event rather than an evolving dynamic system.

Bank balance sheet

- The bank's balance sheet consists of the asset side and liability side.
- There are both inside assets and outside assets and inside liabilities and outside liabilities.
- Assets must equal liabilities, hence equity makes up for the difference.
- Can we see how to make a network out of this?

BANK i BALANCE SHEET			
ASSETS		Liabilities	
Outside assets	c_i	Outside liabilities	b_i
Inside assets	p_{ji}	Inside Liabilities	p_{ij}
		Equity	w_i

Figure: Bank balance sheet for Eisenberg and Noe (2001)

Systemic Risk and ABMs

- Nier et al. (2007) extend the Eisenberg and Noe model to study how shocks propagate over multiple rounds.
- The model effectively functions as an agent-based model (ABM) but is not explicitly labeled as one.
- Each bank is represented as a balance sheet that updates based on a clearing rule.
- Monte Carlo simulations explore how changes in structural parameters affect network dynamics.

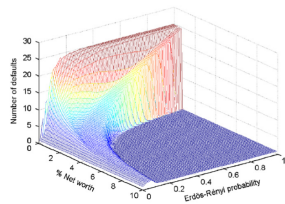
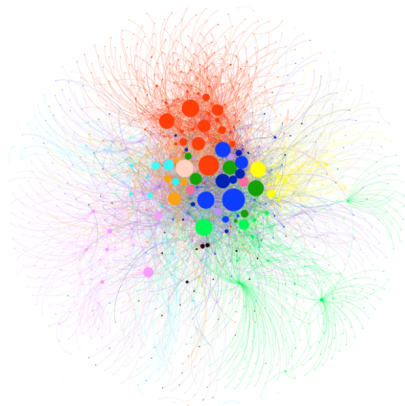


Figure: Number of defaults as a function of net worth and connectivity probability. Source: Nier et al. (2007)

Going from Qualitative to Quantitative



Sources: ECB supervisory data and ECB calculations. Notes: The node size captures the weighted in-degree of interconnectedness. The node colour represents the country of origin. The thickness of the links reflects the value of the exposures in € billions. The colour of the links refers to the country of the target, thus also capturing the borrower's perspective.

Figure: Euro area interbank network of large exposures. Source: Covi et al. (2019)

Assenza et al. (2015): Model overview

As a concrete example, consider the model of Assenza et al. (2015).

Agents

- Households (Worker & Capitalists)
- firms producing consumption goods (C-firms)
- firms producing capital goods (K-firms)
- banks

Markets

- Labour (search & matching)
- C goods (search & matching)
- K goods (search & matching)
- Credit (no search & matching)
- Deposits (no search & matching)

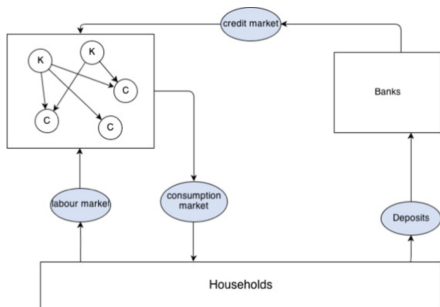


Figure: An overview of Assenza et al. (2015).

Households

- **Labour:** Workers supply one unit of labour inelastically; unemployed workers visit Z_e firms randomly and accept the going wage
- **Income:** Each **capitalist** owns one firm and receives a share of its profits; workers earn wage w
- **Human wealth:** Households form an adaptive estimate of permanent income (exponential moving average of past income)
- **Consumption budget:** Human wealth + a fraction χ of financial wealth
- **Shopping:** Visit Z_c random C-firms, rank by price, buy from cheapest first
- **Involuntary saving:** If goods are unavailable, unspent budget accumulates as deposits

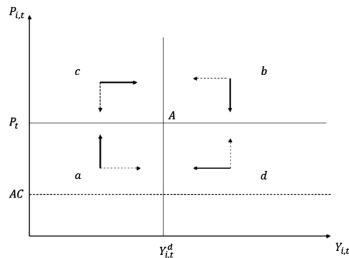
Adaptive Price-Quantity Setting

Firms do not know their demand curve. They observe two signals, **inventories** ($\Delta_{i,t}$) and **relative price** ($P_{i,t}$ vs. $\bar{P}_t$), and adjust *either* price or quantity:

- Quantity adjustment:

$$Y_{i,t+1}^e = \begin{cases} Y_{i,t} + \rho(-\Delta_{i,t}) & \text{if } \Delta_{i,t} \leq 0 \text{ and } P_{i,t} \geq \bar{P}_t, \\ Y_{i,t} - \rho\Delta_{i,t} & \text{if } \Delta_{i,t} > 0 \text{ and } P_{i,t} < \bar{P}_t, \end{cases}$$

- Price adjustment: $P_{i,t+1} = \begin{cases} P_{i,t}(1 + \eta_{i,t+1}) & \text{if } \Delta_{i,t} \leq 0 \text{ and } P_{i,t} < \bar{P}_t, \\ P_{i,t}(1 - \eta_{i,t+1}) & \text{if } \Delta_{i,t} > 0 \text{ and } P_{i,t} \geq \bar{P}_t, \end{cases}$



Both C-firms and K-firms use this mechanism. No firm optimises; each follows a simple heuristic that maps two observable signals to one adjustment.

Figure: Price-quantity setting. (Assenza et al., 2015)

Production and Supply Chain

Consumption goods (C-firms)

- Leontief technology:
 $\hat{Y}_{i,t} = \min(\alpha N_{i,t}, \kappa K_{i,t})$
- Uses both **labour** and **capital**
- Capital depreciates; must be replaced by purchasing from K-firms
- If labour or capital unavailable
⇒ involuntary underproduction

Capital goods (K-firms)

- Linear technology: $Y_{j,t} = \alpha N_{j,t}$
(labour only)
- Upstream supplier: output is a durable, storable input for C-firms
- Same adaptive price-quantity mechanism

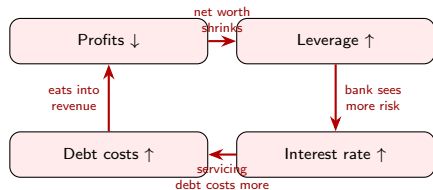
Key point

- Two-sector supply chain creates **interdependence**: C-firms depend on K-firms for capital; K-firms depend on C-firm investment demand

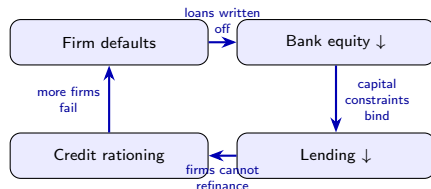
Credit and Financial Accelerator

- Firms borrow to cover the **financing gap**: production costs that exceed internal funds
- The bank sets interest rates based on **leverage** $\lambda = \text{debt}/\text{equity}$
- The bank **rations credit** when expected losses exceed its risk tolerance

Financial Accelerator



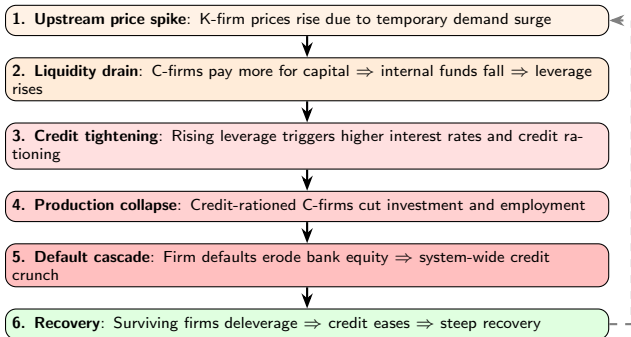
Bank Capital Channel



These loops amplify small shocks into large swings and generate instability **endogenously**.

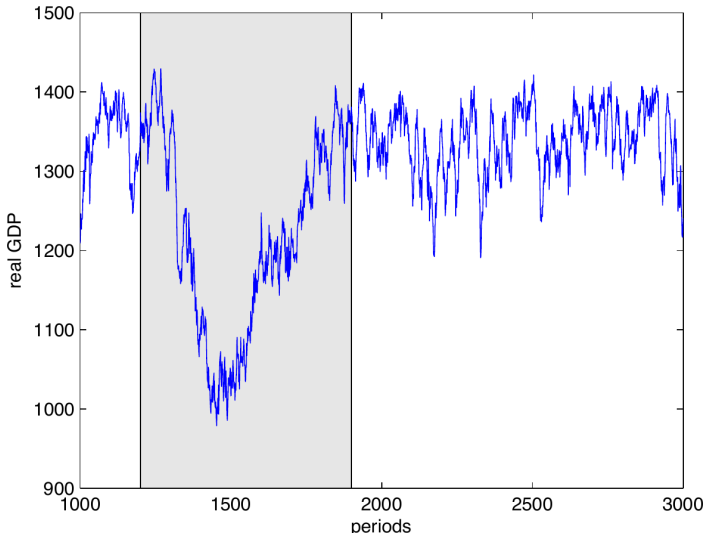
Emergent Crises

How does the model generate crises **without exogenous shocks**?



Key insight: No exogenous shock is needed; the supply chain and credit feedback loops are sufficient.

Simulation Results



Validation

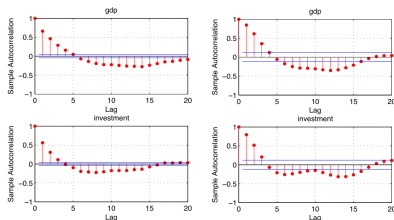


Figure: Autocorrelations: model (left) vs. data (right). (Assenza et al., 2015)

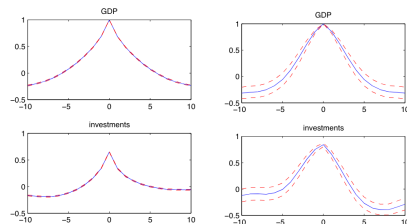


Figure: Correlations: model (left) vs. data (right). (Assenza et al., 2015)

The model reproduces key empirical regularities, persistence (high autocorrelation) and co-movement patterns, without relying on exogenous shocks.

Assenza et al. (2015) summary

Model Structure:

- Two-sector economy:
 - Upstream: Capital goods producers
 - Downstream: Consumption goods firms
- Adaptive behavior for price/quantity decisions
- Banking sector with credit constraints

Key Findings:

- Business cycles emerge without external shocks
- Reproduces empirical time series properties
- Crises occur from sectoral interactions

Crisis Mechanism:

- Spike in capital goods prices
- Consumption firms increase borrowing
- Credit rationing emerges when debt is high
- Production contracts, causing recession
- Recovery follows deleveraging

Implications:

- Upstream-downstream dynamics matter
- Debt cycles drive business cycles
- Adaptive behavior creates persistence

Overview

- Agent based models (ABMs) are simulation-based modelling approach where agents interact.
- We covered some of the general properties that ABMs have.
- We looked at examples of ABMs in microeconomics, finance, and macroeconomics.
- We walked through the Assenza et al. (2015) model as a concrete macro ABM example.
- Next time: quantitative ABMs calibrated to real data, and calibration techniques.
- For now a little demonstration with *BeforeIT.jl* and *ABCCredit.jl* packages which are really well developed Julia packages to get you working with qualitative and quantitative ABMs.

Questions?

Bibliography I

- T. Assenza, D. Delli Gatti, and J. Grazzini. Emergent dynamics of a macroeconomic agent based model with capital and credit. *Journal of Economic Dynamics and Control*, 50:5–28, Jan. 2015.
- R. L. Axtell and J. D. Farmer. Agent-Based Modeling in Economics and Finance: Past, Present, and Future. *Journal of Economic Literature*, Forthcoming.
- W. A. Brock and C. H. Hommes. A Rational Route to Randomness. *Econometrica*, 65(5):1059–1095, 1997. Publisher: Wiley, Econometric Society.
- G. Covi, M. Montagna, and G. Torri. Economic shocks and contagion in the euro area banking sector: a new micro-structural approach. Technical report, May 2019.
- L. Eisenberg and T. H. Noe. Systemic Risk in Financial Systems. *Management Science*, 47(2):236–249, 2001. Publisher: INFORMS.
- I. Gilboa, A. Postlewaite, and D. Schmeidler. The complexity of the consumer problem. *Research in Economics*, 75(1):96–103, Mar. 2021.

Bibliography II

- A. G. Haldane and A. E. Turrell. An interdisciplinary model for macroeconomics. *Oxford Review of Economic Policy*, 34(1-2):219–251, Jan. 2018.
- E. Nier, J. Yang, T. Yorulmazer, and A. Alentorn. Network models and financial stability. *Journal of Economic Dynamics and Control*, 31(6): 2033–2060, June 2007.
- S. Poledna. Canvas: A canadian behavioral agent-based model. Technical report, International Institute for Applied Systems Analysis (IIASA), 2023.
- T. C. Schelling. Models of Segregation. *The American Economic Review*, 59 (2):488–493, 1969. Publisher: American Economic Association.